CLIENT REPORT

Home Depot vs Lowe's: Search Relevance Discoveries

An LLM-judged relevance benchmark of Home Depot and Lowe's on-site search across three query datasets and 900+ graded product judgments — showing where each engine wins, breaks, and loses shoppers.

Prepared for

Home Improvement Retail Search Benchmark

Prepared by an LLM-judged search-evaluation harness

Disclaimer. *A quick, exploratory analysis produced for rapid personal research only. It is nowhere near complete, has not been verified, and must not be cited or referenced. The approach itself is sound and can be made reliable with retailer API access, real query logs (actual search terms), and higher volume.*

AT A GLANCE

Synonyms favor Lowe's (48 vs 30%). Core relevance is a tie (0.91 / 0.88). Both break oppositely on industry specs, and both fail to route project queries to the project pages they already have.

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Executive summary, key findings with evidence, and a scored-terms appendix.

Executive Summary

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THE HEADLINE

Home Depot and Lowe's are near-tied on core relevance, but they fail in opposite, fixable ways. Lowe's wins on synonyms; Home Depot edges project search; and both mishandle industry specs — Home Depot ignores them, Lowe's over-reacts to them.

We evaluated live on-site search on **homedepot.com** and **lowes.com** across three query datasets: (1) **synonym/category** equivalence, (2) **dimensions** — measurement notation and the "#" spec gradations, and (3) **pro/job-function** project and trade queries. Results were captured by bounded, same-origin browser reads (no crawling or scraping); result counts read deterministically; and each returned product graded 0–3 for relevance by a local LLM judge, scored as NDCG. The three headline numbers are at right.

DIMENSIONS NDCG
0.91 / 0.88
Home Depot vs Lowe's — near tie on core relevance.

SYNONYM MATCH
30 / 48%
Lowe's returns consistent results for more equivalent terms.

PROJECT NDCG
0.83 / 0.78
Home Depot edges project and trade queries.

What This Answers

- Which engine returns more relevant results, and where does each break?
- Do the engines understand industry specs ("#57 stone", "#4 rebar") and notation ("1/2-in")?
- Which is better for natural-language project shopping ("build a deck")?

Key Findings

FINDING 01

Industry specs are the biggest gap

The "#" gradations (#57 stone, #4 rebar, #7 drill bit) are hard catalog specs. **Home Depot ignores them** — it returns generic in-category products and never surfaces the true spec. **Lowe's over-reacts** — it treats the "#" as a different query and often derails to the wrong category entirely. Neither reliably finds the exact spec'd item.

FINDING 02

Notation fails oppositely

Home Depot breaks on **hyphenation**: "1/2-in screw" drops about 70% of the results of "1/2 in screw" — a recall cliff. Lowe's breaks on **unit words**: "10 foot cable" returns a fraction of "10 ft cable". Relevance of what each keeps stays high, so these are recall failures, not ranking failures.

FINDING 03

Project intent isn't routed to project pages

Both retailers **publish curated project / job-function pages** (Home Depot's "build a deck" project guide), but search does **not recognize the project intent and route to them** — it returns a generic product list instead. The content already exists; search simply does not connect the query to it. Home Depot also monetizes "install a…" as paid services; both fail bare trade names ("electrician", "landscaper" return **zero products**).

BUSINESS IMPLICATION

These are not vague "search feels off" complaints — each is a specific, reproducible query behavior with a clear fix. Spec parsing and notation normalization are self-contained engine changes; synonym coverage is a mapping problem. All three convert directly into found products, fewer dead-ends, and larger baskets.

Evidence Snapshot

Query (retailer)	What we observed
#57 stone (Lowe's)	Window blinds and a quartz sink; "57 stone" returns real #57 aggregate.
#7 drill bit (Lowe's)	Magnetic drill presses; "7 mm drill bit" returns real twist bits.
#4 rebar vs 4 rebar (HD)	Near-identical sets — the "#" spec is ignored.
1/2-in screw (HD)	Drops about 70% vs "1/2 in screw" (hyphen recall cliff).
replace a kitchen faucet (Lowe's)	5 results, a stray wire listing; HD returns real faucets.

Appendix A — Relevance Judge

Every returned product was graded 0–3 for relevance to the query intent by **Meta Llama 3.2 3B** (model `llama3.2:3b`), run **locally** via Ollama and orchestrated by the Cortex evaluation service at **temperature 0** (deterministic; re-runs are stable). Grades: 3 = perfect match, 2 = relevant, 1 = marginal, 0 = irrelevant. Per-query **NDCG** (0–1) discounts by rank, so a mis-ordered or partly-irrelevant top set scores lower. Every grade is a stored, inspectable trace of the exact prompt and model output. The judge was validated against a hand-labelled gold set (12/13) and two notation defects were fixed before scoring. A "—" below means the query returned no products (or was not captured) on that retailer. The synonym dataset is scored by result-count consistency, not LLM NDCG, so it is not listed here.

Appendix B — Search Terms and NDCG Scores

Search term	Home Depot	Lowe's
Dimensions — notation & hash (32 terms)		
1/2 in screw	1.00	0.95
3/8 in bolt	0.86	0.80
#8 wood screw	1.00	1.00
3/4 in pvc pipe	1.00	1.00
1/2 in copper pipe	0.94	1.00
12 gauge wire	0.98	0.79
10 ft cable	0.79	0.75
10 mm socket	0.86	0.73
#4 rebar	0.98	0.99
#7 drill bit	0.95	0.80
#57 stone	0.79	0.94
#40 roller chain	0.89	0.82
#15 felt	0.97	0.98
4 rebar	0.77	0.96
1/2 in rebar	0.98	1.00
7 mm drill bit	0.80	1.00
7 in drill bit	0.93	0.84
57 stone	0.79	0.72
40 mm chain	0.69	0.79
15 in felt	0.89	0.99
8 in wood screw	0.99	0.95
8 mm wood screw	0.88	0.80
1/2in screw	1.00	0.85
1/2" screw	0.97	0.84
0.5 in screw	1.00	1.00
1/2-in screw	1.00	0.85
no 8 wood screw	0.91	1.00
10' cable	—	—
10 foot cable	—	0.80
120 in cable	—	0.64
10mm socket	—	0.71
15 lb felt	—	0.94
Pro / job-function (20 terms)		
build a deck	0.66	0.48
hang drywall	0.68	0.71
unclog a drain	0.82	0.90
install a ceiling fan	—	0.73
tile a shower	0.96	0.95
paint a room	0.72	0.69
install a toilet	—	—
build a raised garden bed	0.93	0.92
replace a kitchen faucet	0.84	0.00
fix a leaky pipe	0.85	0.97
pour a concrete slab	0.85	0.97
insulate an attic	0.66	0.76

Search term	Home Depot	Lowe's
hang an interior door	0.91	0.97
electrician	—	—
plumber tools	0.82	0.79
welder	0.88	1.00
carpenter tools	0.76	0.79
hvac technician	0.90	0.78
landscaper	—	—
painter tools	1.00	0.91